

DP Physics 11 Lesson Plan

Important links -- [Course Materials by Topic](#) -- [Guides](#) -- [Old DP Physics exams \(SL and HL\)](#) -- IA Resources (TBA)

Topics Covered: [Topic 1](#), [A](#), [D.1](#), [B.5](#), [D.2](#), D.3-- **Current Topic:** -- Next Topic: D (HL)

Term 2, 2026			
W2	<i>Monday Jan 12, 2026</i> Today we went over the Circular Motion + Gravitation Quiz from before winter break.	<i>Tuesday Jan 13, 2026</i> Today we began our next unit of study. This unit will include the SL content from B.5, D.2, and D.3. We will then complete the HL content from Topic D. We used the “Balloons and Static Electricity” PhET to demonstrate the different methods of charging, as well as the ‘charge law’. We ended with slide #17.	<i>Friday Jan 16, 2026</i> Today we continued with B.5 and got to slide #31.
W1	<i>Tuesday Jan 20, 2026</i> Today we began class experimenting with the Van de Graaf Generator. It was a shocking experience! We then continued looking at various aspects of series and parallel circuits. We used this “Resistivity” PhET to help visualize resistance. We got to slide #52.		<i>Thursday Jan 22, 2026</i> Today we continued looking at different aspects of circuits. We used the “Circuit Construction Kit: DC” PhET to demonstrate a number of concepts and to support the equations we’re using. We got to slide #64.
W2	<i>Monday Jan 26, 2026</i> Mr. Knight is away today. You will have the block to work on a series of circuit	<i>Tuesday Jan 27, 2026</i> Today we continued with Topic B.5 and got to slide #75.	<i>Friday Jan 30, 2026</i> Today we continued with Topic B.5 and got to slide #94.

	building activities (worksheet) that will help reinforce the equations for voltage, current, and resistance for series, parallel, and combination circuits.		
W1	<i>Tuesday Feb 3, 2026</i> Today we spent the first half of class conducting an experiment to verify Ohm's Law. Your homework is to complete the worksheet and process your results. The second half class was spent learning about internal resistance as we finished the slides on Topic B.5. We had a few minutes left, so we also began Topic E.2 which is about "Electric Fields". We got to slide #10.	<i>Tuesday Feb 10, 2026</i> Today we began class looking at the "Charges and Fields" PhET which helps visualize how electric fields are generated. It also can be helpful in showing how e-fields behave around monopoles, dipoles, and charged parallel plates. We continued with the notes and got to slide #42.	<i>Thursday Feb 5, 2026</i> Today we began with a discussion of the UBC Physics Olympics, which are taking place at UBC on Saturday Feb 28. Email Victoria at victoriah26@stratfordhall.ca if you are interested in joining our school's team! We then spent some time going over the results of the Ohm's Law lab from Tuesday. I showed you how to process the data in Google Sheets, and then how to graph and analyze it using LoggerPro. We spent the remainder of the class on the class notes. We got to slide #15.
W2	<i>Monday Feb 9, 2026</i> Today we continued with notes and got to slide #30.		<i>Friday Feb 13, 2026</i> Pro-D -- no school

W1	<i>Tuesday Feb 17, 2026</i> Today we completed a lab where we determined the internal resistance of a D-cell battery. We then finish Topic D.2.		<i>Thursday Feb 19, 2026</i> Today we began D.3. We got to slide #18.
W2	<i>Monday Feb 23, 2026</i> Today, we continued with Topic D.3 and learned how moving charges generate magnetic fields. We got to slide #27.	<i>Tuesday Feb 24, 2026</i> Today, we carried on with D.3 notes. We finished SL content from Topic D today, and got to slide #47. The remainder of the slides are practice problems. We will complete them next class as part of our preparation for the unit test next week.	<i>Friday Feb 27, 2026</i> Finish D.3 slides + review for unit test
W1	<i>Tuesday Mar 3, 2026</i> Topic D SL Unit Test		<i>Thursday Mar 5, 2026</i> Today we went over the Topic D SL unit test. Next, we will cover the AHL (Additional High-Level) content for Topic D. This will take us a couple weeks and bring us past spring break. The next area of study will either be Topic B (SL) or Topic C (SL), depending on timing.
W2	<i>Monday Mar 9, 2026</i> Today we began Topic D (HL). We got to slide #25 in the slide deck.	<i>Tuesday Mar 10, 2026</i> Today we continued with Topic D HL content and got to slide #39.	<i>Friday Mar 13, 2026</i> Today we continued with Topic D HL content and got to slide #60. Happy Spring Break!

November/December 2025

W2	<i>Monday Nov 3, 2025 (double block due to assembly on Friday)</i> Today we finished Topic A.2.1 and took a step aside to look at A.3. We will finish A.2.2 and A.2.3 after we're done. We got to slide #7. We also looked at the Energy Skate Park PhET which helps demonstrate concepts like the conservation of energy.		<i>Tuesday Nov 4, 2025</i> Today we continued Topic A.3 and looked at the concept of "work", Hooke's Law, and the elastic potential energy. We ended on slide #19.
	<i>Monday Nov 10 - Friday Nov 14</i> November Break -- no school!		
W1	<i>Tuesday Nov 18, 2025</i> Today we continued with Topic A.3 and primarily focused on the conservation of mechanical energy. We just about finished the topic, but some tech issues made us run out of time. We will finish A.3 next time.		<i>Thursday Nov 20, 2025</i> Today we finished Topic A.3 with a discussion of power and efficiency. When then continued with Topic A.2.2 "Momentum and impulse" and got to slide #14.
W2	<i>Monday Nov 24, 2025</i> Today we continued Topic A.2.2, but did not finish it. We got to slide #26.	<i>Tuesday Nov 25, 2025</i> Today we finished Topic A.2.2. It will be up to each of you to be your own review for the unit test. Please see Topic A Course Material folder for PDF study guides. You are all welcome to attend my Study Centre hours in the MYP on Thursday.	<i>Friday Nov 28, 2025</i> Topic A Unit Test
W1	<i>Tuesday Dec 2, 2025</i> Today we went over the Topic A Unit Test. This took all of C-block.		<i>Thursday Dec 4, 2025</i>

	In D block, we began Topic A.2.3 “Circular motion”. We used the “Ladybug Revolution” PhET to discover some key concepts, and then started the slides. We got to slide #14.		Today we continued on with Topic A.2.3 circular motion. We got to slide #30.
W2	<i>Monday Dec 8, 2025</i> Today we finished A.2.3. We will begin Topic D.1 next class.	<i>Tuesday Dec 9, 2025</i> Today we began Topic D.1 “Gravitational fields” (SL). We began things with a demonstration of the gravitational field using a large black cloth that we stretched over a large circular hoop and rolled balls of various size/mass across it. We then went to the slides and defined Newton’s Universal Law of Gravitation. We got to slide #10.	<i>Friday Dec 12, 2025</i> Today we continued with D.1 and discussed the concept of the gravitational field strength “g”. We got to slide #25.
Mini-Term	<i>Wednesday Dec 17, 2025 (PM)</i> Finish D.1 Unit Test Review		<i>Thursday Dec 18, 2025</i> Topic A.2.3 + D.1 (Circular Motion + Gravitation) Unit Test
	<i>Monday Dec 22 - Friday Jan 9</i> Winter Break -- no school!		

October 2025		
W1	<i>Tuesday Sep 30, 2025</i> National Day for Truth and Reconciliation -- no school	<i>Thursday Oct 2, 2025</i> Topic 1 Unit Test

W2	<i>Monday Oct 6, 2025</i> Today we went over the Topic 1 Unit Test. It was our first test of the year, and many of you likely did not perform as well as you had hoped. It's okay -- this was meant as a "check-in" for you and I both to see if you are keeping up in the course. There will be plenty more opportunities to improve your mark moving forward. If you want to improve, now is the time to increase your effort both in and out of class. Reminder: My Study Centre day is Thursday in the MYP.	<i>Tuesday Oct 7, 2025</i> Today we began Topic A "Space, time and motion" with an introduction into Topic A.1 "Kinematics". We looked at displacement, velocity, and acceleration, and got to slide #21.	<i>Friday Oct 10, 2025</i> Today we continued Topic A.1 and learned about the kinematic equations, also called the "suvat" equations in IB. We got to slide #43.
W1	<i>Tuesday Oct 14, 2025</i> Today in class we continued practicing the kinematic equations. We also learned how to calculate and determine the meaning of the slope and area under the curve for position-time, velocity-time, and acceleration-time graphs. We ended on slide #55 In our second block, we did a "Freefall of Gravity" lab. In the lab, you recorded the position of a falling object over time. Your homework is to graph this data and determine the acceleration due to gravity "g" <u>graphically</u>.		<i>Thursday Oct 16, 2025</i> Today I went through the process of analyzing the data from the "Freefall of Gravity" lab from last class. I showed you a number of ways of determining "g" graphically, including through linearization. The purpose of this was to show you how in-depth a good IA analysis section should be.
W2	<i>Monday Oct 20, 2025</i> Today we continued with Topic A.1 and learned about projectile motion. We ended on slide #79. I asked you to solve the problem on that slide as homework.	<i>Tuesday Oct 21, 2025</i> Today we completed Topic A.1 and moved on with Topic A.2.1, which is about forces. We got to slide #6.	<i>Friday Oct 24, 2025</i> Pro-D Day -- no school
W1	<i>Tuesday Oct 28, 2025</i>		<i>Thursday Oct 30, 2025</i>

	Today we carried on with Topic A.2. We discussed different types of forces, Free Body Diagrams, and Newton's first law of motion. We got to slide #25.	Today we continued looking at forces (A.2) and completed a number of practice problems as a class. We got to slide #37.
--	--	---

June/Aug/Sept 2025			
W2	<i>Tuesday Jun 10, 2025</i> Today we introduced the course and DP science in general. Class notes here. We will use this page (known as the "Organizing Doc") to manage important class info. This includes the daily happenings in class, assessment scheduling, and important course documents like the Physics Guide, class notes, study guides, old exam files, etc.		<i>Thursday Jun 12, 2025</i> Today we completed a few more Paper 1A type questions (multiple choice). We then did mini-lessons on the factor-labeling method and basic trig functions. See links for the worksheets I asked you to complete. We ended class with this xkcd comic.
	<i>Friday Aug 29, 2025</i> Today I was unable to attend class, so I had you write a "baseline" test to determine what you remember from last year and how you are at solving novel problems.		
W1	<i>Tuesday Sep 2, 2025</i> New student orientation -- no classes		<i>Thursday Sep 4, 2025</i> Today we welcomed everyone back from summer break, and then we talked a little about last year's results. We also went over the Paper 1A test from last week.
W2	<i>Monday Sep 8, 2025</i> Begin Topic 1 "Measurements and uncertainties"	<i>Tuesday Sep 9, 2025</i> Continue Topic 1 "Measurements and uncertainties"	<i>Friday Sep 12, 2025</i> DP Retreat -- no classes

	Mr. Knight away at Grade 9 camp	Mr. Knight away at Grade 9 camp	
W1	<i>Tuesday Sep 16, 2025</i> Last week was a bit of a bust due the issue with coverage while I was away at camp. Thank you to those of you who pre-read the Topic 1 slides. Today we went through the Topic 1.1 slides on “Measurements in physics” and began the Topic 2.2 slides on “Uncertainties and errors”. We ended with the dartboard analogy of precision vs accuracy on slide #11.		<i>Thursday Sep 18, 2025</i> Today we finished Topic 1.2 with a discussion on different ways to account for error propagation (algebraically and graphically).
W2	<i>Monday Sep 22, 2025</i> Today we completed the Measurements and Uncertainties series of lab activities where we practiced using various lab equipment, recording measurements and errors, and propagating errors through our calculations. We watched a couple videos to help us learn how to use the Vernier calipers and the micrometer screwgauge.	<i>Tuesday Sep 23, 2025</i> Today we began Topic 1.3 “Vectors and scalars”. We used the “Vector Addition HYPERLINK” "https://phet.colorado.edu/sims/html/vector-addition/latest/vector-addition_all.html" HYPERLINK "https://phet.colorado.edu/sims/html/vector-addition/latest/vector-addition_all.html" PhET to illustrate many of the details of vectors. We covered most of the content and ended with slide #20. We will finish the slide deck next class and begin studying for our first unit test.	<i>Friday Sep 26, 2025</i> Today we finished Topic 1. We then discussed what to expect on the upcoming Topic 1 Unit Test next Thursday. For review material, see the PDF Study Guides in this resource folder. Additionally, please complete this set of Paper 1B practice (old exam) questions. Printed copies of the questions are in a red folder in the DP library. Digital copies of the questions and markschemes are in the folder linked above.

Month Template			
W1	<i>Tuesday</i> <TBA>		<i>Thursday</i> <TBA>
W2	<i>Monday</i>	<i>Tuesday</i>	<i>Friday</i>

	<TBA>	<TBA>	<TBA>
W1	<i>Tuesday</i>		<i>Thursday</i>
	<TBA>		<TBA>
W2	<i>Monday</i>	<i>Tuesday</i>	<i>Friday</i>
	<TBA>	<TBA>	<TBA>
W1	<i>Tuesday</i>		<i>Thursday</i>
	<TBA>		<TBA>
W2	<i>Monday</i>	<i>Tuesday</i>	<i>Friday</i>
	<TBA>	<TBA>	<TBA>